Assissment #4

Meta-analysis of the microbial communities in wastewater and natural water reservoirs

Inna Kucherova https://github.com/inna-tsymb/metagenomics/tree/main/lecture_04

1. Research question and data

The main objective of current study wass to determine the presence of fundamental structural and composition differences between anthropogenic wastewater (municipal, hospital, sludge) and natural water reservoirs (rivers). The key hypothesis was that natural river environments demonstrate different microbial diversity and have unique groups of specialists which are fundamentally differ from anthropogenic wastewater.

Data from four different cohort studies were combined for meta-analysis: municipal watewater plants (Shultz, 2017), activated sludge (Chu, 2017), hospital wastewater (Rowe, 2017), river (Lekunberri, 2018), and pond (Chopyk, 2020). The initial dataset contained 250 samples from fractional metagenomic sequencing (MetaPhlAn4). To avoid statistical bias due to dominance of one cohort, a normilized mode was applied: the Shultz sample size was reduced to n=20. Data were filtered at the species level (prevalence ≥10%, average abundance ≥0.01%) and normalized using CLR transformation.

2. Main results and conclusions

The complex analysis consistetly confirmed the hypothesis that water source type is the dominant factor in the fotmation of the microbiome, which causes complete replacement of ecological niches.

Ecological comparison: analysis showed that although the difference in alpha-diversity between anthropogenic and natural groups wasnn't statistically significant (p=0.281), there are strong marker species. Eg, *Flavobacterium tructae* and *Limnohabitans sp. Rim47* are markers of natural environments, while *Aliarcobacter cryaerophilus* and *Nitrospira sp. ND1* are markers of anthropogenic environments.

MWAS: linear regression (ANOVA) detected that 735 of 746 species (98.5%) in the normalized cohort were significatly associated with the source of origin (FDR<0.05). Pairwise comparisons shewed that the most contrasting pair was pond compared to treatment plants wastewater, where 89.0% of species differ.

Meta-analysis and sensitivity testing: fixed-effect meta-analysis confirmed statistical significance for 502 of 684 species (73.4%). Also, there was indicated high heterogeneity between cohorts (median I^2 = 85.5%), which is expected for environmental metagenomes. Sensitive meta-analysis on fully balanced samples (n=60(showed increasing in the proportion of significant species to 76.7%, which confirms that sample size imbalance can mask ecological signals.

Table 1. Top differentiated species

Species	Associated environment	Effect (CLR, pooled)	Heterogenity (I^2)
Limnohabitans sp., Polynucleobacter spp.	**Natural (river/pond)**	**Hign enrichment**	**~85-98%**
f__Burkholderiaceae	**Specific**	**-5.39**	**98.5%**
p__Actinobacteria	**Specific**	**-8.77**	**99.4%**

Meta-analysis confirmed previous findings. Natural waters dominated by freshwater microorganisms (eg, Limnohabitans, Polynucleobacter), which totally absent in anthropogenic environments. Extremely high discrimination between groups 9>89% of species differ) confirms that wastewater microbiome doesn't simply displace or mix with the natural one. However, both communities face a hard ecological barrier: the survival of bacteria dictated by environmental conditions, so these microbiomes are practically incapable of coexisting or replacing each other outside their ecologocal nishes.

3. Comparison with published data

To verify the findings, the results were compared with the study by Wu et al., 2019.

Consistency of results: Wu et al. demonstrated that wastewater microbiomes differ globally from natural freshwater systems and form specific core. That's fully consistent with current results, where typical wastewater

treatment plant bacterias were identified as anthropogenic markers, while natural water maker species exclusively dominated in river and pond.

Comparison of strategies and additional analyses: while this analysis based on fractional sequencing (MetaPhlAn4) and regression (ANOVA) on CLR-transformed data, Wu et al. used 16s rRNA sequencing. The authors also analyzed how communities form using a null model approach. This showed that deterministic selection is the main factor in engineered treatment systems. As a result, engineered conditions prevent natural freshwater microbes from surviving. When wastewater microbes enter clean water, they die and cannot replace the native oligotrophs. This leads to an "all or nothing" pattern of colonization.

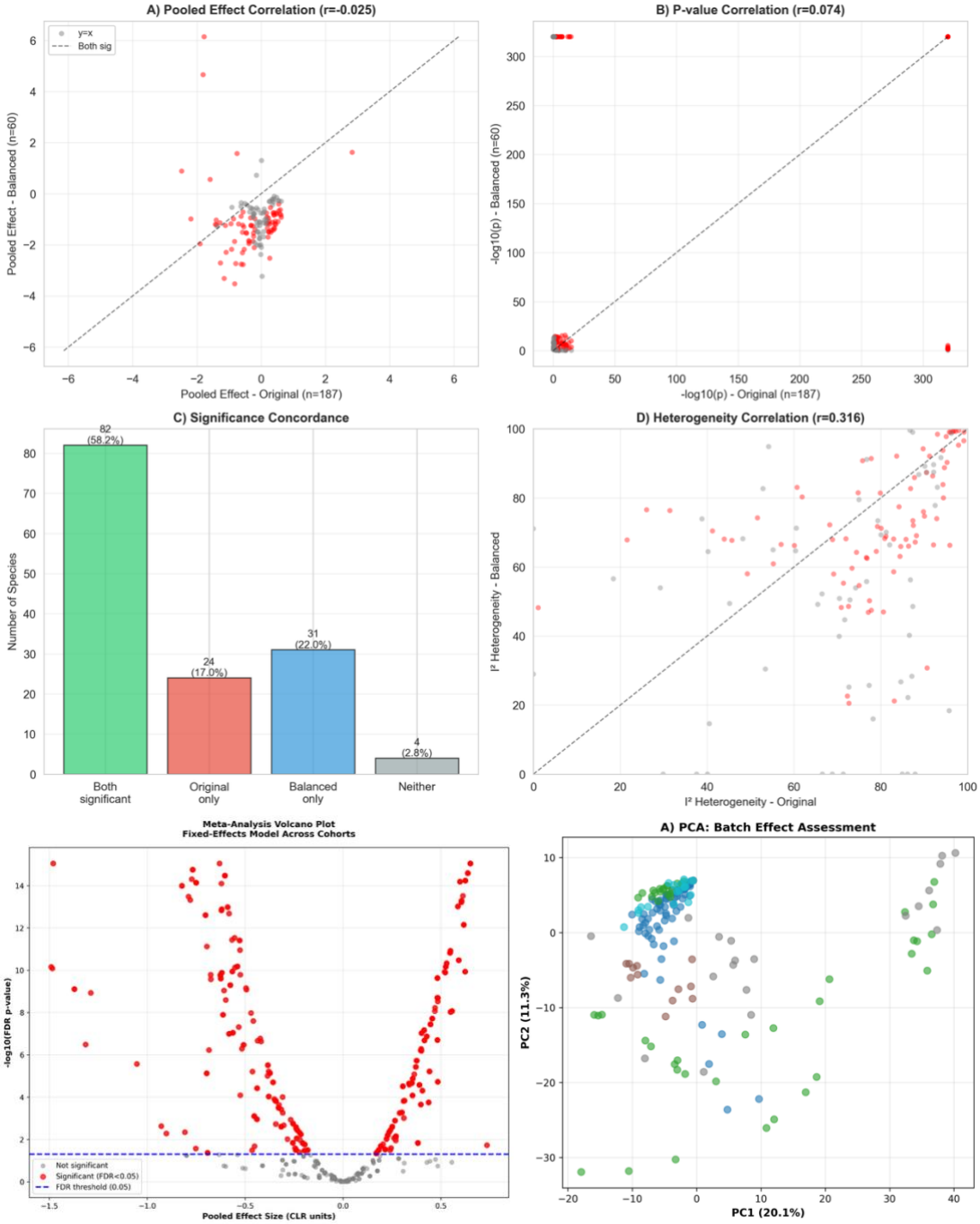

Fig. 1. Comprehensive results of meta-analysis and sensitivity testing of the balanced model. (Bottom right) Evaluation of batch effects using PCA demonstrates clear spatial clustering of samples depending on cohort membership. (Bottom left) A volcano plot for the fixed-effects meta-analysis shows a massive distribution of statistically significant species (red dots, FDR < 0.05). (Top panels) Comparison of metrics between the original (n=187) and balanced (n=60) meta-analysis designs. There is moderate overall concordance (58.2% of types remain significant in both models), but virtually no correlation of pooled effect sizes (r = -0.025), confirming the significant impact of sample size imbalance on the estimation of global effects.

Literature

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